

AI-Powered Enterprise Email Intelligence

MailMind

Technical Architecture & Developer Documentation

|         |                                                            |
|---------|------------------------------------------------------------|
| Version | v2.0.0                                                     |
| Date    | 2025                                                       |
| Stack   | FastAPI · Next.js 15 · LangGraph · Azure OpenAI · Supabase |
| Team    | Radiant CodeX                                              |
| License | MIT                                                        |

This document provides a comprehensive technical reference for the MailMind codebase — covering system architecture, file structure, per-file API documentation, LangGraph pipeline design, database schema, and deployment infrastructure.

# Executive Summary

MailMind is an AI-powered enterprise email triage and drafting platform built for busy executives who receive hundreds of emails per day. It automatically scores every incoming email across five contextual axes, extracts calendar commitments, detects conflicts, retrieves relevant precedents, and generates personalised draft replies — all in under 3 seconds.

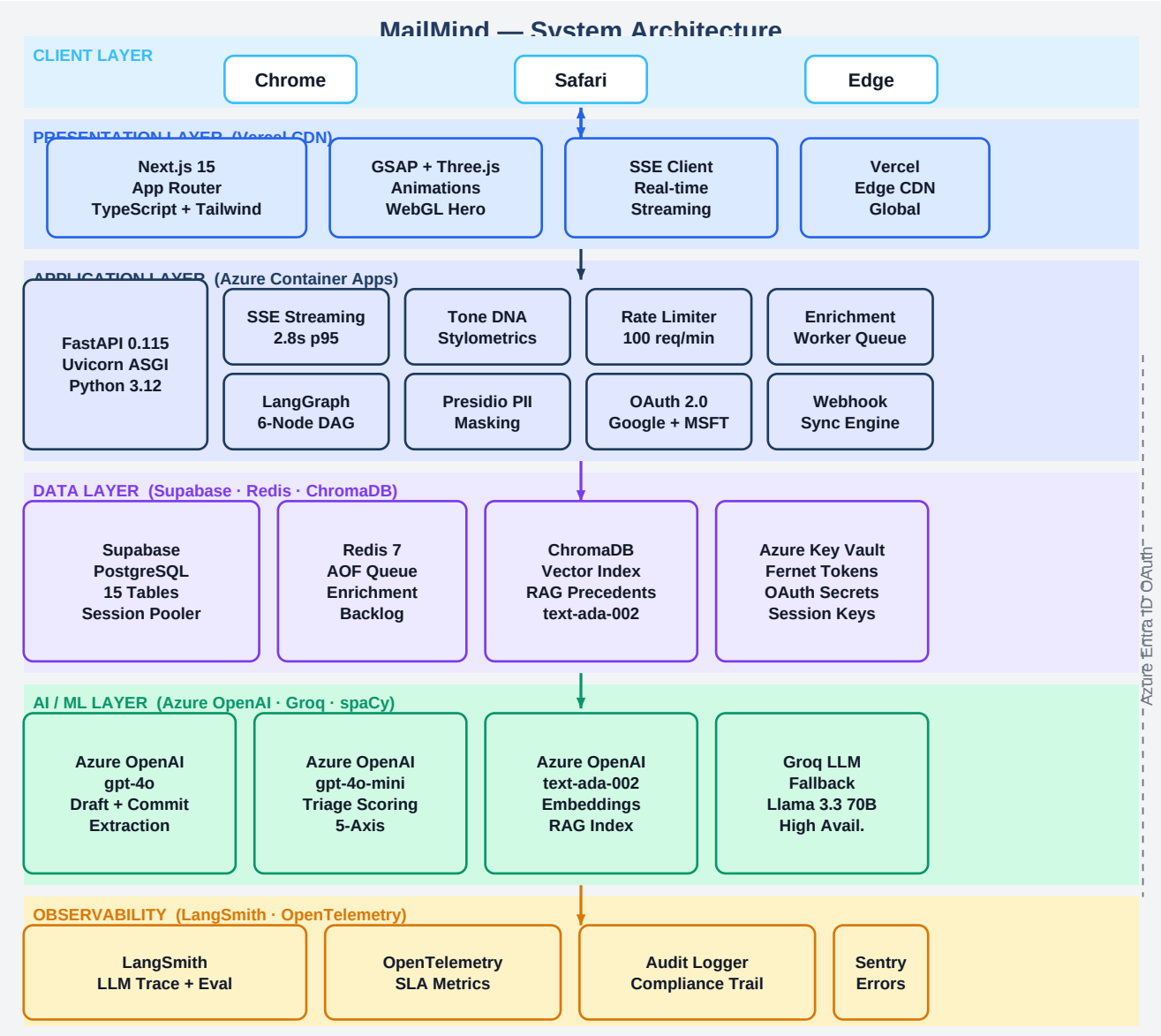
| Key Feature           | Description                                                                               | Tech                 |
|-----------------------|-------------------------------------------------------------------------------------------|----------------------|
| 5-Axis Triage         | Scores emails on deadline urgency, sender authority, sentiment, action type, thread decay | GPT-4o-mini          |
| Tone DNA              | Learns writing style from 50+ sent emails (8 stylometric features)                        | spaCy + custom NLP   |
| RAG Drafting          | Retrieves 3 similar past emails and injects context into draft generation                 | ChromaDB + Ada-002   |
| PII Masking           | Presidio + spaCy reversible token masking before LLM calls                                | Microsoft Presidio   |
| Commitment Extraction | Extracts action items with deadlines + calendar conflict detection                        | GPT-4o structured    |
| Mailbox Mirror        | Server-side delta sync (cursor-based) + Graph/Gmail webhooks                              | Graph API + Pub/Sub  |
| Multi-Account         | UUID-based user identity, multiple Gmail/Outlook accounts per user                        | PostgreSQL           |
| Demo Login            | One-click seeded demo with 10 emails, full triage cache, VP persona                       | seed_demo_account.py |

## Technology Stack

| Backend                                                                                                                                                        | Frontend                                                                                                                                         | AI / ML                                                                                                                                                                                       | Infrastructure                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• FastAPI 0.115</li><li>• Python 3.12</li><li>• LangGraph 0.2</li><li>• LangChain 0.3</li><li>• SQLAlchemy 2.0</li></ul> | <ul style="list-style-type: none"><li>• Next.js 15</li><li>• TypeScript 5</li><li>• Tailwind CSS 3</li><li>• GSAP 3</li><li>• Three.js</li></ul> | <ul style="list-style-type: none"><li>• Azure OpenAI gpt-4o</li><li>• gpt-4o-mini (triage)</li><li>• text-embedding-ada-002</li><li>• Groq Llama 3.3</li><li>• spaCy en_core_web_sm</li></ul> | <ul style="list-style-type: none"><li>• Azure Container Apps</li><li>• Azure Key Vault</li><li>• Supabase PostgreSQL</li><li>• Redis 7 AOF</li><li>• ChromaDB (local)</li></ul> |

# System Architecture

MailMind follows a layered architecture: Next.js frontend on Vercel, FastAPI backend on Azure Container Apps, and separate data/AI/observability layers. All traffic is HTTPS; OAuth tokens are Fernet-AES-128 encrypted at rest.



# Cloud Infrastructure

All Azure resources live in a single resource group. Container Apps handle auto-scaling; Key Vault provides secrets management. GitHub Actions builds and deploys on every push to main.

